

# Regular polygons and torsional rigidity

## A course in three lectures

Detailed explanations accompanying the torsion manuscript

1 October 2026

These notes follow the manuscript from the vertex Hessian to a computer-assisted local maximality proof. Each lecture is also available as a separate PDF. The second lecture develops a regularity proof at the regular polygon and proves the resulting quadratic Galerkin Hessian rate. Uniform estimates on neighborhoods of perturbed polygons remain a separate question. The numerical implementation described is the maintained `MaxTorsionValidation/` source snapshot of 13 September 2026.

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## Part I

# From the shape Hessian to its eigenvalues

## 1 The question and the coordinates

**Aim.** Starting from Laurain's distributed second shape derivative, obtain the two eigenvalues of every Fourier block of the vertex Hessian. A suggested 90-minute session spends 20 minutes on vertex calculus, 25 on the local blocks, 30 on Fourier reduction, and 15 on the checks and exercises. The prerequisite is the weak Dirichlet problem and the meaning of a second shape derivative; no interval arithmetic is used in this lecture.

For a polygon  $P$ , let

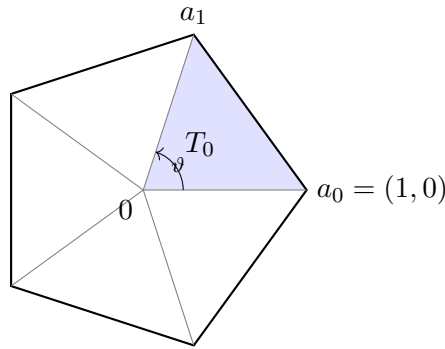
$$-\Delta u = 1 \quad \text{in } P, \quad u \in H_0^1(P), \quad J(P) = \frac{1}{2} \int_P u = \frac{1}{2} \int_P |\nabla u|^2.$$

Our objective is  $\mathcal{F}(P) = J(P)/A(P)^2$ , where  $A(P) = |P|$ . Conventional torsional rigidity for this PDE is  $2J$ . Multiplying the objective by two changes eigenvalue magnitudes, but not their signs. The dilation laws  $u_{tP}(tx) = t^2 u_P(x)$  and  $J(tP) = t^4 J(P)$  explain the power  $A^2$  in the denominator.

Fix the regular polygon of circumradius one. Throughout the course,

$$\vartheta = \frac{2\pi}{n}, \quad a_j = (\cos(j\vartheta), \sin(j\vartheta)), \quad T_j = \text{conv}\{0, a_j, a_{j+1}\}.$$

Indices are taken modulo  $n$ . The same  $\vartheta$  is used everywhere; we do not give separate names to its half-angle, its multiples, or its sine and cosine. The rotation through  $j\vartheta$  is  $R_j$ . The vector of vertex coordinates is interleaved as  $(x_0, y_0, x_1, y_1, \dots)$ .



Let  $\phi_j$  be the continuous affine hat on this coarse fan, one at  $a_j$  and zero at its other vertices, including the centre. A vertex perturbation  $q = (q_j)_{j=0}^{n-1}$ ,  $q_j \in \mathbb{R}^2$ , is extended into the polygon by

$$\theta_q(x) = \sum_j \phi_j(x) q_j.$$

This formula says exactly how points inside the polygon move: the map  $x \mapsto x + s\theta_q(x)$  moves the boundary vertices as prescribed and fixes the centre. For sufficiently small  $s$ , it is a bi-Lipschitz map. Its gradient is constant on each fan triangle. This last fact will be used in every subsequent calculation.

## 2 From a shape derivative to a vertex Hessian

For each vertex,  $U_i = (U_i^1, U_i^2)$  collects the material derivatives of  $u$  under displacements of that vertex in the two coordinate directions. The superscript is a component, not a derivative order. Differentiating the pulled-back weak equation gives, for scalar  $v \in H_0^1(P)$ ,

$$\begin{aligned} \int_P DU_i \nabla v &= \int_P -(\nabla u \cdot \nabla v) \nabla \phi_i + (\nabla \phi_i \cdot \nabla v) \nabla u \\ &\quad + (\nabla \phi_i \cdot \nabla u) \nabla v + v \nabla \phi_i. \end{aligned} \quad (1)$$

For the direction  $q$ , the scalar material derivative is  $u_q = \sum_i q_i \cdot U_i$ . All these functions have zero Dirichlet trace on the reference polygon: the boundary condition was pulled back before being differentiated.

We assume Laurain's distributed formula [5, Proposition 14]. Its planar reduction is manuscript equation 8:

$$\begin{aligned} (D^2 J)_{ij} &= \int_P DU_i DU_j^T \\ &\quad + \int_P \left( \frac{1}{2} |\nabla u|^2 + u \right) (\nabla \phi_i \otimes \nabla \phi_j - \nabla \phi_j \otimes \nabla \phi_i) \\ &\quad - \int_P (\nabla \phi_i \cdot \nabla \phi_j) (\nabla u \otimes \nabla u). \end{aligned} \quad (2)$$

This is a  $2 \times 2$  block. For example its Gram contribution in component  $(p, q)$  is  $\int_P \nabla U_i^p \cdot \nabla U_j^q$ .

Here is the algebra behind the reduction. For any three planar vectors  $p, r, g$ ,

$$\begin{aligned} (r \cdot g)p \otimes g + (p \cdot g)g \otimes r - (p \cdot g)r \otimes g - (r \cdot g)g \otimes p \\ = |g|^2(p \otimes r - r \otimes p). \end{aligned}$$

Checking the two diagonal entries gives zero on both sides; checking one off-diagonal entry gives the other by antisymmetry. Substitute  $p = \nabla \phi_i$ ,  $r = \nabla \phi_j$ ,  $g = \nabla u$  in Laurain's integrand. The remaining terms proportional to  $u$  give the skew expression in (2). This is a two-dimensional simplification; it should not be used unchanged in dimension three.

There are two useful structural checks. First, individual blocks need not be symmetric, but  $(D^2 J)_{ji} = ((D^2 J)_{ij})^T$ . Second, the last two terms are local: the hats have overlapping gradients only for identical or neighboring vertices. The Gram term is the only generally dense part.

### 2.1 The area correction must be differentiated too

The shoelace formula gives

$$A = \frac{1}{2} \sum_j (x_j y_{j+1} - y_j x_{j+1}), \quad \nabla_{a_i} A = \frac{1}{2} (y_{i+1} - y_{i-1}, x_{i-1} - x_{i+1}).$$

The only nonzero mixed second derivatives connect neighboring vertices:  $A_{x_i y_{i+1}} = 1/2$  and  $A_{y_i x_{i+1}} = -1/2$ , together with their symmetric counterparts. At the regular polygon,  $A = n \sin \vartheta / 2$  and  $\nabla_{a_i} A = \sin \vartheta a_i$ .

By the ordinary product rule,

$$\begin{aligned} D^2 \mathcal{F} &= A^{-2} D^2 J - 2JA^{-3} D^2 A \\ &\quad - 2A^{-3} (\nabla J \otimes \nabla A + \nabla A \otimes \nabla J) + 6JA^{-4} \nabla A \otimes \nabla A. \end{aligned} \quad (3)$$

Dihedral symmetry makes  $\nabla_{a_i} J$  a common multiple of  $a_i$ . Euler's identity for the degree-four functional  $J$  then gives

$$\nabla J = \frac{2J}{A} \nabla A, \quad D\mathcal{F}(P_n) = 0.$$

Substitution into (3) yields

$$D^2\mathcal{F} = A^{-2}D^2J - 2JA^{-3}D^2A - 2JA^{-4}\nabla A \otimes \nabla A. \quad (4)$$

This formula is valid at the regular polygon. On the area-tangent space,  $D^2J - (2J/A)D^2A$  has the same signs as  $D^2\mathcal{F}$ . One must not inspect  $D^2J$  alone: radial dilation increases  $J$ .

### 3 Compute the local part on two triangles

The hat gradients at vertex zero are

$$\nabla\phi_0 = (1, -\cot\vartheta) \text{ on } T_0, \quad \nabla\phi_0 = (1, \cot\vartheta) \text{ on } T_{-1},$$

and zero elsewhere. Also,  $\nabla\phi_1 = (0, 1/\sin\vartheta)$  on  $T_0$  and  $\nabla\phi_{-1} = (0, -1/\sin\vartheta)$  on  $T_{-1}$ . These identities follow by solving the three nodal interpolation equations, not by differentiating a trigonometric interpolant.

Only three sector integrals are needed:

$$X = \int_{T_0} u_x^2, \quad Y = \int_{T_0} u_y^2, \quad Z = \int_{T_0} u_x u_y.$$

Reflection across each radial side makes  $\partial_n u = 0$  there. On the polygon side  $u = 0$ . Integration by parts within  $T_0$  therefore gives

$$X + Y = \int_{T_0} u = \frac{2J}{n}, \quad \int_{T_0} (\tfrac{1}{2}|\nabla u|^2 + u) = \frac{3J}{n}.$$

Reflection across the sector bisector diagonalizes the sector energy tensor in the bisector basis. In Cartesian coordinates this means

$$(Y - X) \sin\vartheta + 2Z \cos\vartheta = 0. \quad (5)$$

In particular,  $-X \sin\vartheta + Z \cos\vartheta = -J \sin\vartheta/n$ . This identity is the cancellation to remember; storing  $Z$  separately is unnecessary for the final symbol.

Write  $L_j$  for the first block row of the local, non-Gram part of  $D^2J$ . Substitution of the hat gradients into (2) gives

$$\begin{aligned} L_0 &= -\frac{2}{\sin^2\vartheta} \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix}, \\ L_1 &= \frac{\cos\vartheta}{\sin^2\vartheta} \begin{pmatrix} X & Z \\ Z & Y \end{pmatrix} - \frac{3J}{n \sin\vartheta} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \\ L_{-1} &= \frac{\cos\vartheta}{\sin^2\vartheta} \begin{pmatrix} X & -Z \\ -Z & Y \end{pmatrix} + \frac{3J}{n \sin\vartheta} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \end{aligned} \quad (6)$$

All other  $L_j$  vanish. To check the sign in  $L_1$ , note that  $\nabla\phi_0 \cdot \nabla\phi_1 = -\cos\vartheta/\sin^2\vartheta$ , whereas  $\nabla\phi_0 \otimes \nabla\phi_1 - \nabla\phi_1 \otimes \nabla\phi_0$  is the negative of the displayed rotation matrix divided by  $\sin\vartheta$ .

### 3.1 What remains unknown in the sector integrals?

If  $\Delta_n$  denotes the difference of the energies parallel and perpendicular to the sector bisector, rotation of that diagonal tensor gives

$$X = \frac{J}{n} + \frac{\Delta_n}{2} \cos \vartheta, \quad Y = \frac{J}{n} - \frac{\Delta_n}{2} \cos \vartheta, \quad Z = \frac{\Delta_n}{2} \sin \vartheta.$$

Symmetry alone does not determine  $\Delta_n$ . The manuscript's mixed Dirichlet–Neumann torsion problem provides an analytic characterization of it; the fixed-mesh certificate does not need that separate computation. For the square,  $X = Y = J/4$  even though  $Z$  need not vanish. The sector is not symmetric under reflection in the Cartesian axes individually.

## 4 Why radial and tangential coordinates reveal the spectrum

At vertex  $j$ , the radial unit vector is  $a_j$  and the tangential unit vector is  $(-\sin(j\vartheta), \cos(j\vartheta))$ . They are the columns of  $R_j$ . With  $R = \text{diag}(R_0, \dots, R_{n-1})$ , set

$$H = R^T D^2 \mathcal{F} R.$$

This is an orthogonal change of coordinates, so eigenvalues are unchanged. In these coordinates rotation of the polygon merely shifts the vertex index. Consequently  $H_{ij} = C_{j-i}$ , where  $C_{-j} = C_j^T$ .

The block-circulant reduction is due to Tee [9] and is used in this polygonal setting by Bogosel and Bucur [2]. It can be proved in one line. For a sequence  $v_j = e^{ikj\vartheta} w$ ,  $w \in \mathbb{C}^2$ ,

$$(Hv)_i = \sum_j C_{j-i} e^{ikj\vartheta} w = e^{iki\vartheta} \left( \sum_{j=0}^{n-1} e^{ikj\vartheta} C_j \right) w.$$

Thus the  $k$ th Fourier subspace carries the  $2 \times 2$  matrix

$$B_k = \sum_{j=0}^{n-1} e^{ikj\vartheta} C_j. \tag{7}$$

The normalized discrete Fourier transform is unitary; the multiset of all  $2n$  Hessian eigenvalues is the union of the spectra of these  $n$  blocks.

Symmetry gives  $B_k^* = B_k$ . Reflection in the axis through vertex zero acts as  $\text{diag}(1, -1)$  on radial/tangential components and sends  $j$  to  $-j$ . The diagonal sequences are even and the off-diagonal sequence is odd. Therefore

$$B_k = \begin{pmatrix} \alpha_k & i\gamma_k \\ -i\gamma_k & \beta_k \end{pmatrix}, \quad \mu_k^\pm = \frac{\alpha_k + \beta_k \pm \sqrt{(\alpha_k - \beta_k)^2 + 4\gamma_k^2}}{2}. \tag{8}$$

The parameters  $\alpha_k, \beta_k, \gamma_k$  are real. These are manuscript equations (39)–(40). The sign of the imaginary entry depends on the Fourier convention; the convention in (7) is fixed throughout these notes.

### 4.1 The explicit cancellation with area

The radial/tangential area symbol is

$$\sin \vartheta \cos(k\vartheta) I - i \cos \vartheta \sin(k\vartheta) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

In the first row, the local blocks become  $L_j R_j$ , since  $R_0 = I$ . Multiply the three matrices in (6) by  $R_j$ , use (5), and subtract the area contribution in (4). The result is

$$\begin{aligned} A^{-2} \sum_{j=-1}^1 e^{ikj\vartheta} L_j R_j - 2JA^{-3} \left[ \sin \vartheta \cos(k\vartheta) I - i \cos \vartheta \sin(k\vartheta) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] \\ = -\frac{2(1 - \cos(k\vartheta))}{A^2 \sin^2 \vartheta} \begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix}. \end{aligned} \quad (9)$$

The explicit off-diagonal term disappears completely. The rank-one term  $\nabla A \otimes \nabla A$  in (4) acts only on the constant radial sequence, hence only on mode zero.

For clarity, introduce the scalar rotated components already used in the manuscript:

$$U_{j,r} = \cos(j\vartheta)U_j^1 + \sin(j\vartheta)U_j^2, \quad U_{j,t} = -\sin(j\vartheta)U_j^1 + \cos(j\vartheta)U_j^2, \quad a(v, w) = \int_P \nabla v \cdot \nabla w.$$

The three genuinely nonlocal quantities are

$$\begin{aligned} G_k^{rr} &= \sum_j \cos(kj\vartheta) a(U_0^1, U_{j,r}), \\ G_k^{tt} &= \sum_j \cos(kj\vartheta) a(U_0^2, U_{j,t}), \\ G_k^{rt} &= \sum_j \sin(kj\vartheta) a(U_0^1, U_{j,t}). \end{aligned}$$

For every nonzero mode, (9) now yields

$$\boxed{\begin{aligned} \alpha_k &= A^{-2} \left[ G_k^{rr} - \frac{2(1 - \cos(k\vartheta))X}{\sin^2 \vartheta} \right], \\ \beta_k &= A^{-2} \left[ G_k^{tt} - \frac{2(1 - \cos(k\vartheta))Y}{\sin^2 \vartheta} \right], \\ \gamma_k &= A^{-2} G_k^{rt}. \end{aligned}} \quad (10)$$

These are the requested Hessian eigenvalue formulas together with (8). Their trace contains the explicit penalty  $-4(1 - \cos(k\vartheta))J/(nA^2 \sin^2 \vartheta)$ . Sector anisotropy cancels from that explicit penalty; it does not cancel from the state-dependent Gram terms.

## 5 Exact zero modes and the local maximum criterion

At a critical point of a similarity-invariant function, differentiating invariance puts the four infinitesimal similarity directions in the Hessian kernel. The mode bookkeeping matters:

| motion          | radial/tangential coordinates           | Fourier location |
|-----------------|-----------------------------------------|------------------|
| scaling         | (1, 0) at every vertex                  | $k = 0$          |
| rotation        | (0, 1) at every vertex                  | $k = 0$          |
| $x$ translation | $(\cos(j\vartheta), -\sin(j\vartheta))$ | $k = 1, n - 1$   |
| $y$ translation | $(\sin(j\vartheta), \cos(j\vartheta))$  | $k = 1, n - 1$   |

Thus  $B_0 = 0$ . At  $k = 1$  the complex translation vector is  $(1, i)$ , so  $\alpha_1 = \beta_1 = \gamma_1$ . The other eigenvalue is the trace  $\alpha_1 + \beta_1$ . The conjugate mode  $n - 1$  gives the other real translation and the same nonzero eigenvalue.

Also  $B_{n-k} = \overline{B_k}$ . If  $n$  is even, the Nyquist mode  $k = n/2$  has zero sine sequence and  $\gamma_{n/2} = 0$ . These are exact identities, not quantities to be recognized by a small numerical threshold. For a generic block, strict negativity is equivalent to

$$\alpha_k + \beta_k < 0, \quad \alpha_k \beta_k - \gamma_k^2 > 0.$$

This determinant test excludes translation blocks, whose determinant is exactly zero; there the trace alone is tested.

**Theorem 5.1** (What the numerical calculation must establish). *If every branch outside the four similarity directions is strictly negative, the regular polygon is a strict local maximizer of  $J/A^2$  modulo similarities. Equivalently, it is a strict local maximizer of  $J$  at fixed area, modulo translations and rotations.*

*Proof.* The fan pullback has smoothly parameter-dependent, uniformly elliptic coefficients in a sufficiently small vertex neighborhood. Its inverse solution operator is smooth, so the objective is twice differentiable. Choose a local slice complementary to the four-dimensional similarity orbit. The Hessian is negative definite on the slice; Taylor's theorem gives strict decrease for small nonzero slice displacements. Every nearby orbit meets the slice. Fixing area removes the dilation freedom. This proves a local statement about nearby labelled polygons, not global optimality among all polygons.  $\square$

## 6 The quantities actually passed to the next lecture

One need not assemble the entire Hessian to evaluate its symbol. For  $1 \leq k < n/2$ , the following are normalized vertex displacements:

$$\begin{aligned} r_{k,j}^c &= \sqrt{2/n} \cos(kj\vartheta) (\cos(j\vartheta), \sin(j\vartheta)), & \text{radial, cosine,} \\ t_{k,j}^c &= \sqrt{2/n} \cos(kj\vartheta) (-\sin(j\vartheta), \cos(j\vartheta)), & \text{tangential, cosine,} \\ t_{k,j}^s &= \sqrt{2/n} \sin(kj\vartheta) (-\sin(j\vartheta), \cos(j\vartheta)), & \text{tangential, sine.} \end{aligned}$$

Their pairings give respectively

$$D^2 \mathcal{F}[r_k^c, r_k^c] = \alpha_k, \quad D^2 \mathcal{F}[t_k^c, t_k^c] = \beta_k, \quad D^2 \mathcal{F}[r_k^c, t_k^c] = 0, \quad D^2 \mathcal{F}[r_k^c, t_k^s] = \gamma_k.$$

For the Nyquist cosine directions use  $(-1)^j/\sqrt{n}$  instead of  $\sqrt{2/n} \cos(kj\vartheta)$ . The certificate encloses these real pairings directly; it need not compute separate intervals for  $X, Y, Z$  or the Gram sums.

Finally, covariance gives

$$U_j(x) = R_j U_0(R_j^T x), \quad U_{j,r}(x) = U_0^1(R_j^T x), \quad U_{j,t}(x) = U_0^2(R_j^T x).$$

The distinction between vector rotation and scalar pullback explains the optimized code: torsion and two first-vertex PDEs supply every first material derivative. Later second-lifting solves error estimation.

### Exercises and answers for discussion

**Exercise 6.1.** *Derive  $\nabla J = (2J/A)\nabla A$  without a boundary integral formula.*

*Answer.* Reflection makes each gradient radial; rotation makes their coefficients equal. If  $\nabla_{a_j} J = ca_j$ , Euler's identity gives  $nc = 4J$ . Compare with  $\nabla_{a_j} A = \sin \vartheta a_j$ .

**Exercise 6.2.** *For  $n = 5$ , explain why the target is six negative eigenvalues, even though there are only two representative nonzero modes.*

*Answer.* Modes 1, 4 contribute one physical eigenvalue each; modes 2, 3 contribute two each. Mode zero and the two translations supply four zeros. The total is  $2 + 4 = 6$  negatives out of ten dimensions.

**Exercise 6.3.** *Can a negative computed trace alone certify a generic  $2 \times 2$  block?*

*Answer.* No. A block with eigenvalues  $-2$  and  $1$  has negative trace. A determinant bound or two strictly negative eigenvalue enclosures is also required.

## Part II

# Finite element errors, fluxes, and transmission regularity

## 7 Three errors that must not be confused

**Aim.** Turn a finite element Hessian entry into an enclosure of the continuous entry. Then investigate the transmission regularity behind its quadratic Galerkin convergence. A 90-minute lecture can cover the error identities and first-derivative regularity; the second-derivative argument below is suitable for a further reading session.

The geometry is the same regular polygon and coarse fan as in the first lecture. Write  $a(v, w) = \int_P \nabla v \cdot \nabla w$  and  $\|v\|_a = \|\nabla v\|_{L^2(P)}$ . A displacement  $q$  has the fan extension  $\theta_q = \sum_j q_j \phi_j$ . These are vertex displacements, not functions that can be chosen independently on adjacent triangles. Recall that  $-\Delta u = 1$ ,  $u \in H_0^1(P)$ ,  $\ell(v) = \int_P v$ ,  $J = \frac{1}{2} \int_P u$ , and  $\mathcal{F} = J/A^2$  with  $A = |P|$ .

There are three distinct objects:

| object      | meaning                              | source of the difference |
|-------------|--------------------------------------|--------------------------|
| $u$         | continuous weak solution             | —                        |
| $u_h$       | exact Galerkin solution in $V_h$     | spatial discretization   |
| $\tilde{u}$ | computed coefficient vector in $V_h$ | linear-system solve      |

Interval rounding is a third error, incurred while evaluating quantities in the proof. It is handled by enclosing each arithmetic operation. A small solver residual alone controls neither  $u - u_h$  nor the Hessian error. Conversely, an a priori interpolation theorem does not control the rounding error in a computed residual.

Each coarse triangle is split uniformly into  $m^2$  fine triangles. The mesh has  $nm^2$  triangles and  $1 + nm(m+1)/2$  vertices. Its boundary has  $nm$  vertices. For fixed  $n$ ,  $h = m^{-1}$  is proportional to the maximum fine-element diameter. All rays are unions of element edges, and the coefficients obtained by differentiating the pullback are constant on each fine triangle. A mesh crossing a ray cannot use the same broken interpolation argument without additional analysis. The space  $V_h$  consists of continuous functions that are affine on each fine triangle and zero on the polygon boundary. The exact Galerkin state is characterized by  $a(u_h, v_h) = \ell(v_h)$  for every  $v_h \in V_h$ . It is a mathematical solution of a finite linear system, independently of how accurately a numerical solver finds its coefficients.

## 8 A residual is an error measured in the dual norm

For  $u$  satisfying  $a(u, v) = \ell(v)$  and any conforming candidate  $w$ ,

$$a(u - w, v) = \ell(v) - a(w, v).$$

Taking the supremum over  $\|v\|_a = 1$  gives an equality:

$$\|u - w\|_a = \|\ell - a(w, \cdot)\|_{(H_0^1, a)^*}.$$

The equality does not compute the dual norm. It identifies what a guaranteed majorant must bound.

For torsion, let  $y \in H(\text{div}; P)$  satisfy  $\text{div } y = -1$ . Integration by parts, with the zero trace of  $v$ , gives

$$\ell(v) - a(w, v) = \int_P (y - \nabla w) \cdot \nabla v, \quad \|u - w\|_a \leq \|y - \nabla w\|_{L^2(P)}.$$

This is a guaranteed estimate: it is valid for every admissible  $y$ , not only the minimizer of a flux-fitting problem. It is a functional majorant in the sense discussed by Repin and by the guaranteed finite element literature [8, 7, 6].

A convenient exact construction is

$$y = -x/2 + \text{curl } \psi_h, \quad \text{curl } \psi_h = (\partial_y \psi_h, -\partial_x \psi_h), \quad (11)$$

where  $\psi_h$  is continuous and piecewise affine. In distributions,  $\text{div curl } \psi_h = 0$ . Across an element edge the normal component of the curl is a tangential derivative of the common trace of  $\psi_h$ , so it agrees from both sides. Thus this is a global  $H(\text{div})$  field with precisely the required divergence.

The numerical code chooses  $\psi_h$  by minimizing a quadratic mismatch. Its computed nodal values are simply admissible flux data. An inaccurate fit widens the bound; it does not destroy equilibration. The first part  $-x/2$  is essential: a curl alone would have the wrong divergence.

**Exercise 8.1.** *What happens if the flux is not exactly equilibrated?*

*Answer.* The still-valid estimate is  $\|u - w\|_a \leq \|y - \nabla w\| + C_F \|1 + \text{div } y\|$ , where  $\|v\| \leq C_F \|v\|_a$ . Exact equilibration removes the second term.

## 9 Differentiate the weak problem, not the approximate output

For two vertex directions, use  $x \mapsto x + s\theta_q(x) + t\theta_r(x)$ . The pulled-back coefficient matrix and volume density are

$$\mathbb{A}(s, t) = \det(I + sD\theta_q + tD\theta_r)(I + sD\theta_q + tD\theta_r)^{-1}(I + sD\theta_q + tD\theta_r)^{-T},$$

$$a_{s,t}(v, w) = \int_P \nabla v^T \mathbb{A}(s, t) \nabla w, \quad \ell_{s,t}(v) = \int_P \det(I + sD\theta_q + tD\theta_r) v.$$

Subscripts  $q, r, qr$  on forms mean derivatives at  $s = t = 0$ . In particular,

$$\begin{aligned} \mathbb{A}_q &= (\text{div } \theta_q)I - D\theta_q - D\theta_q^T, \\ \ell_q(v) &= \int_P (\text{div } \theta_q) v, \\ \ell_{qr}(v) &= \int_P [(\text{div } \theta_q)(\text{div } \theta_r) - \text{tr}(D\theta_q D\theta_r)] v. \end{aligned}$$

Differentiating  $a_{s,t}(u_{s,t}, v) = \ell_{s,t}(v)$  gives

$$\begin{aligned} a(u_q, v) &= \ell_q(v) - a_q(u, v), \\ a(u_{qr}, v) &= \ell_{qr}(v) - a_q(u_r, v) - a_r(u_q, v) - a_{qr}(u, v). \end{aligned} \quad (12)$$

All derivatives here are material derivatives on the fixed reference fan. They exist in  $H_0^1$  by differentiability of the uniformly coercive solution operator; that fact alone does not give broken  $H^2$  regularity.

Replace the test space by  $V_h$  and the state by  $u_h$ . The exact discrete first derivatives solve the resulting equations, and  $z_h = u_{h,qr}$  solves the fourth discrete equation. Define the continuous lifting by

$$a(Z_{qr}^h, v) = \ell_{qr}(v) - a_q(u_{h,r}, v) - a_r(u_{h,q}, v) - a_{qr}(u_h, v). \quad (13)$$

Its Galerkin solution is  $z_h$ . The superscript  $h$  on  $Z_{qr}^h$  refers to its data;  $Z_{qr}^h$  itself is not a finite element function.

The following form bounds are finite maxima of matrix operator norms:

$$|a_q(v, w)| \leq C_q \|v\|_a \|w\|_a, \quad |a_r(v, w)| \leq C_r \|v\|_a \|w\|_a, \quad |a_{qr}(v, w)| \leq C_{qr} \|v\|_a \|w\|_a.$$

They transfer errors between the successive equations. No smallness of  $C_q, C_r, C_{qr}$  is assumed.

## 10 Why the second variation admits products of errors

Set  $e = u - u_h$ ,  $e_q = u_q - u_{h,q}$ , and  $e_r = u_r - u_{h,r}$ . Galerkin orthogonality gives the exact identity

$$J - J_h = \frac{1}{2}a(e, e). \quad (14)$$

It holds on nearby polygons after pullback. Differentiating twice means differentiating the form as well as both arguments:

$$(J - J_h)_{qr} = \frac{1}{2}a_{qr}(e, e) + a_q(e_r, e) + a_r(e_q, e) + a(e_q, e_r) + a(e_{qr}, e).$$

Since  $u_{h,qr} \in V_h$ , orthogonality gives  $a(e_{qr}, e) = a(u_{qr}, e)$ . Insert the second equation in (12), tested with  $e$ , and combine the terms. The result is

$$(J - J_h)_{qr} = a(e_q, e_r) - \frac{1}{2}a_{qr}(e, e) + a(Z_{qr}^h - z_h, e). \quad (15)$$

The exact second derivative has disappeared. If

$$\|e\|_a \leq \delta_0, \quad \|e_q\|_a \leq \delta_q, \quad \|e_r\|_a \leq \delta_r, \quad \|Z_{qr}^h - z_h\|_a \leq \eta_{qr},$$

then

$$|(J - J_h)_{qr}| \leq \delta_q \delta_r + \frac{1}{2}C_{qr}\delta_0^2 + \delta_0 \eta_{qr}. \quad (16)$$

This is manuscript equation 64. It is valid on each fixed mesh without an asymptotic regularity assumption.

### 10.1 Could we use only material derivatives?

Yes. The weak equation already gives  $a(u_q, u_r) = \ell_q(u_r) - a_q(u, u_r)$ . It uses no second derivative. But replacing  $u$  by  $u_h$  changes the load, so it does not automatically remove the terms linear in the state error. Direct propagation of  $O(h)$  energy bounds through the Hessian generally leaves an  $O(h)$  bound. The Bogosel–Bucur analysis already exploits a quadratic interpolation product for material derivatives; the leading first-order contribution comes from other data errors, not a failure of Galerkin orthogonality [3, Section 3.1].

In (16), all four errors of order  $h$  instead produce an order- $h^2$  bound. This improvement is due to the cancellation and lifting. Equilibrated fluxes are a way to enclose the energy errors; they are not, by themselves, a theorem of quadratic convergence.

## 11 The computed linear systems also need error bounds

Let  $Kx_* = f$  be the exact interior-node system, with  $K$  symmetric positive definite. A computed vector  $x$  has residual  $f - Kx$ . If  $\lambda_{\min}(K) \geq \underline{\alpha} > 0$ , then

$$\|x - x_*\|_K^2 = (f - Kx)^T K^{-1} (f - Kx) \leq \frac{\|f - Kx\|_2^2}{\underline{\alpha}}.$$

The Euclidean norm of the residual is not the energy norm of the error; the factor  $1/\sqrt{\alpha}$  supplies that conversion.

For the circumradius-one polygon, domain monotonicity and the square give  $\lambda_1(P) \geq \pi^2/2$ . On one triangle,

$$M_T = \frac{|T|}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \succeq \frac{|T|}{12} I.$$

After assembly and restriction to interior nodes,

$$\underline{\alpha} = \frac{\pi^2}{24} \min_{i \text{ interior}} \sum_{T \ni i} |T| \quad \text{is a valid lower bound for } \lambda_{\min}(K). \quad (17)$$

This is explicit and possibly conservative. It contains no numerically estimated inverse and no uncertified eigenvalue computation.

Define the four residual functionals of the computed candidates:

$$\begin{aligned} \rho_0(v) &= \ell(v) - a(\tilde{u}, v), \\ \rho_q(v) &= \ell_q(v) - a_q(\tilde{u}, v) - a(\tilde{u}_q, v), \\ \rho_r(v) &= \ell_r(v) - a_r(\tilde{u}, v) - a(\tilde{u}_r, v), \\ \rho_z(v) &= \ell_{qr}(v) - a_q(\tilde{u}_r, v) - a_r(\tilde{u}_q, v) - a_{qr}(\tilde{u}, v) - a(\tilde{z}, v). \end{aligned}$$

Their vectors are obtained by evaluating them on each interior basis function  $\varphi_i$ :  $(r_\rho)_i = \rho(\varphi_i)$ . For example,  $r_{\rho_0} = f - K\tilde{x}$ .

Because later loads contain earlier unknowns, the transfer is

$$\begin{aligned} \epsilon_0 &= \|r_{\rho_0}\|_2 / \sqrt{\alpha}, \\ \epsilon_q &= \|r_{\rho_q}\|_2 / \sqrt{\alpha} + C_q \epsilon_0, \\ \epsilon_r &= \|r_{\rho_r}\|_2 / \sqrt{\alpha} + C_r \epsilon_0. \end{aligned}$$

These bound the differences between exact discrete solutions and their candidates. Boundary coefficients are set to exactly zero before any residual is assembled. A large penalty imposed by a floating-point solver is not a substitute for this Dirichlet restriction.

### 11.1 Correct the centre before increasing the radius

The energy centre  $\tilde{J}_{\text{corr}} = \ell(\tilde{u}) - a(\tilde{u}, \tilde{u})/2$  satisfies  $J_h - \tilde{J}_{\text{corr}} = \|u_h - \tilde{u}\|_a^2/2$ . For the mixed derivative use

$$\begin{aligned} \tilde{C}_{\text{corr}} &= \ell_{qr}(\tilde{u}) + \ell_q(\tilde{u}_r) - \frac{1}{2} a_{qr}(\tilde{u}, \tilde{u}) - a_q(\tilde{u}, \tilde{u}_r) \\ &\quad + \rho_r(\tilde{u}_q) + \rho_0(\tilde{z}). \end{aligned}$$

Expanding about the exact discrete fields and substituting the residual identities gives

$$\begin{aligned} J_{h,qr} - \tilde{C}_{\text{corr}} &= \rho_z(u_h - \tilde{u}) - \frac{1}{2} a_{qr}(u_h - \tilde{u}, u_h - \tilde{u}) \\ &\quad + a(u_{h,q} - \tilde{u}_q, u_{h,r} - \tilde{u}_r). \end{aligned}$$

Every first-order material-solve term has cancelled. Consequently

$$|J_{h,qr} - \tilde{C}_{\text{corr}}| \leq E_{\text{alg}} := \frac{\|r_{\rho_z}\|_2}{\sqrt{\alpha}} \epsilon_0 + \frac{1}{2} C_{qr} \epsilon_0^2 + \epsilon_q \epsilon_r.$$

The lifting solve error need not be given another separate name: its residual appears directly in this expression.

## 12 Four fluxes and the final continuous enclosure

Choose four continuous affine potentials. The state flux is (11); the first-derivative flux is  $-\theta_q + \text{curl } \psi_{q,h}$ , with divergence  $-\text{div } \theta_q$ . For the second lifting use

$$-(\text{div } \theta_q)\theta_r + D\theta_q\theta_r + \text{curl } \psi_{z,h}.$$

Indeed,

$$\text{div } ((\text{div } \theta_q)\theta_r - D\theta_q\theta_r) = (\text{div } \theta_q)(\text{div } \theta_r) - \text{tr}(D\theta_q D\theta_r).$$

Its normal trace is continuous across fan edges: the gradient jump of a continuous affine field is a vector tensor the edge normal, and the two normal jump contributions cancel. This is a distributional identity, not only a trianglewise identity.

The four mismatch norms are

$$\begin{aligned}\Phi_0 &= \| -x/2 + \text{curl } \psi_{0,h} - \nabla \tilde{u} \|, \\ \Phi_q &= \| -\theta_q + \text{curl } \psi_{q,h} - \nabla \tilde{u}_q - \mathbb{A}_q \nabla \tilde{u} \|, \\ \Phi_r &= \| -\theta_r + \text{curl } \psi_{r,h} - \nabla \tilde{u}_r - \mathbb{A}_r \nabla \tilde{u} \|, \\ \Phi_z &= \| -(\text{div } \theta_q)\theta_r + D\theta_q\theta_r + \text{curl } \psi_{z,h} - \nabla \tilde{z} \\ &\quad - \mathbb{A}_q \nabla \tilde{u}_r - \mathbb{A}_r \nabla \tilde{u}_q - \mathbb{A}_{qr} \nabla \tilde{u} \|.\end{aligned}$$

All unmarked norms in these formulas are  $L^2(P)$  norms of vector fields. Residual bounds, load-error transfer, and best approximation give

$$\begin{aligned}\hat{\delta}_0 &= \Phi_0, \\ \hat{\delta}_q &= \Phi_q + C_q \Phi_0 + \epsilon_q, & \hat{\delta}_r &= \Phi_r + C_r \Phi_0 + \epsilon_r, \\ \hat{\eta}_{qr} &= \Phi_z + C_q \epsilon_r + C_r \epsilon_q + C_{qr} \epsilon_0.\end{aligned}$$

For example,  $\Phi_0$  bounds the error to  $\tilde{u}$ ; best approximation then also bounds the error to  $u_h$ . In the lifting bound,  $\tilde{z} \in V_h$  can replace  $z_h$  before best approximation is used. This explains why an extra lifting algebraic error is not added there.

For a nonzero Fourier pairing,  $A_q = A_r = 0$ . Define

$$B_J = \hat{\delta}_q \hat{\delta}_r + \frac{1}{2} C_{qr} \hat{\delta}_0^2 + \hat{\delta}_0 \hat{\eta}_{qr}.$$

The entry centre and radius are

$$\begin{aligned}\tilde{F}_{qr} &= \frac{\tilde{C}_{\text{corr}}}{A^2} - \frac{2\tilde{J}_{\text{corr}} A_{qr}}{A^3}, \\ R_F &= \frac{B_J + E_{\text{alg}}}{A^2} + \frac{|A_{qr}|}{A^3} (\hat{\delta}_0^2 + \epsilon_0^2), \\ |D^2 \mathcal{F}[q, r] - \tilde{F}_{qr}| &\leq R_F.\end{aligned}\tag{18}$$

Every term has a job:  $B_J$  is the continuous discretization error,  $E_{\text{alg}}$  is the error in the discrete centre, and the last term accounts for the area-normalization coefficient multiplying  $J$ . If the evaluated centre is an interval, its arithmetic radius is retained.

## 13 Regularity proof: integrate the vertex-zero load by parts

This section develops the proposed Bogosel–Bucur route. We will prove broken regularity of first and second material derivatives *at the regular polygon*. The positive shift is justified by a compatible trace lifting and reflection, followed by the Dirichlet polygonal shift theorem. The

manuscript now proves this fixed-polygon result in Proposition 6.2 and its quadratic Hessian rate in Theorem 6.3. Uniformity over perturbed polygons remains a separate question.

For  $c = 1, 2$ , put the weak equation for  $U_0^c$  in the form

$$\begin{aligned} a(U_0^c, v) &= \int_P G_{0,c}(u) \cdot \nabla v + \int_P (\partial_c \phi_0) v, \\ G_{0,c}(u) &= -(\partial_c \phi_0) \nabla u + (\partial_c u) \nabla \phi_0 + (\nabla \phi_0 \cdot \nabla u) e_c. \end{aligned} \quad (19)$$

On a sector the hat gradient is constant. Since  $\Delta u = -1$ ,

$$\operatorname{div} G_{0,c}(u) = \partial_c \phi_0 + 2 \nabla \phi_0 \cdot \nabla \partial_c u.$$

The volume load after integration by parts is therefore

$$-2 \nabla \phi_0 \cdot \nabla \partial_c u \in L^2(P). \quad (20)$$

It is supported on  $T_{-1} \cup T_0$ . The term  $\partial_c \phi_0$  has cancelled; keeping it would give an incorrect strong formulation.

Let  $S_j = \{ra_j : 0 < r < 1\}$  and let  $\nu_j = (-\sin(j\vartheta), \cos(j\vartheta))$  point from  $T_{j-1}$  into  $T_j$ . The trace convention is  $[w]_j = w|_{T_j} - w|_{T_{j-1}}$ . The contribution of the two boundary integrals is  $-[G_{0,c}(u)]_j \cdot \nu_j$ . Continuity of  $\phi_0$  implies  $[\nabla \phi_0]_j$  is normal to  $S_j$ . Because  $u \in H^2(P)$ , its gradient has a common trace there. Substituting into (19) gives the especially simple ray density

$$-[G_{0,c}(u)]_j \cdot \nu_j = -([\nabla \phi_0]_j \cdot \nu_j) \partial_c u|_{S_j}. \quad (21)$$

Only  $S_{-1}, S_0, S_1$  occur. Their coefficients are

$$[\nabla \phi_0]_0 \cdot \nu_0 = -2 \cot \vartheta, \quad [\nabla \phi_0]_{\pm 1} \cdot \nu_{\pm 1} = \frac{1}{\sin \vartheta}.$$

Thus the complete distributional equation is

$$-\Delta U_0^c = -2 \nabla \phi_0 \cdot \nabla \partial_c u + 2 \cot \vartheta (\partial_c u)|_{S_0} \delta_{S_0} - \frac{(\partial_c u)|_{S_1} \delta_{S_1} + (\partial_c u)|_{S_{-1}} \delta_{S_{-1}}}{\sin \vartheta}. \quad (22)$$

Here  $g\delta_S$  means the functional  $v \mapsto \int_S gv \, ds$ . No point masses occur: we integrated bounded piecewise coefficients times  $H^1$  gradient fields once, and the singular terms are edge measures.

### 13.1 Why the ray densities have the right endpoint behavior

For fixed  $n$ , convex-polygon Dirichlet regularity gives

$$u \in H^{2+\sigma}(P), \quad 0 < \sigma < \min \left\{ \frac{1}{2}, \frac{\pi}{\pi - 2\pi/n} - 1 \right\},$$

after decreasing  $\sigma$  if necessary. Possible resonant logarithmic corner terms are harmless below this strict exponent threshold. In particular  $\nabla u$  is continuous up to each corner. Its tangential derivatives on the two adjacent Dirichlet edges are zero; the tangents are independent, so  $\nabla u(a_j) = 0$ . At the centre, rotational symmetry implies  $\nabla u(0) = 0$ .

On  $S_j$ , reflection gives zero normal derivative, hence

$$\nabla u(ra_j) = a_j \partial_r u(r, 0).$$

Thus the densities in (22) are precisely rotated multiples of the same radial derivative. The trace theorem gives  $H^{1/2+\sigma}$  regularity on each ray, and the values at  $r = 0, 1$  vanish. Their extensions by zero across the centre belong to the same fractional space for sufficiently small  $\sigma > 0$ . Endpoint values here are justified by the extra positive regularity; bare  $H^{1/2}$  membership would not by itself justify a pointwise endpoint argument.

### 13.2 The mixed-boundary ingredient, with its hypotheses visible

**Lemma 13.1** (A ray load with vanishing endpoints). *Fix a regular polygon. Let  $g \in H^{1/2+\sigma}(0, 1)$  with  $g(0) = g(1) = 0$  and  $\sigma > 0$  sufficiently small. The solution  $W \in H_0^1(P)$  of  $a(W, v) = \int_{S_0} gv$  belongs to  $H^{2+\sigma'}$  on each side of the ray and on every fan sector, for some  $\sigma' > 0$  that may be smaller than  $\sigma$ . The same assertion holds for any rotated ray.*

*Proof.* The line load is continuous on  $H_0^1(P)$  by the trace theorem, so Lax–Milgram gives  $W$ . Its data and domain are invariant under reflection across the line through  $S_0$ , hence  $W$  is even. Restrict to the upper half-polygon. The boundary conditions are Dirichlet zero on the polygon boundary and outward Neumann data  $g/2$  on the positive part of the cut, zero on its negative part. The factor  $1/2$  follows by doubling the weak integral over the half-polygon.

To justify a positive shift, lift the nonzero Neumann data before reflecting. Choose  $0 < \sigma' < \min\{\sigma, 1/2, 2/(n-2)\}$ . Prescribe zero value on the entire half-polygon boundary, outward normal derivative  $g/2$  on the loaded cut, and zero normal derivative on all remaining sides. Theorem 1 of Bernardi–Dauge–Maday [1], with  $p = 2$ ,  $s = 2 + \sigma'$ , and two traces, gives a lifting  $v_0 \in H^{2+\sigma'}$ . The corner compatibility conditions are precisely agreement of values and first gradients here. They hold because the prescribed value and both endpoint normal derivatives are zero. Zero extension through the centre preserves the required  $H^{1/2+\sigma'}$  regularity of the cut data.

The difference  $W - v_0$  has homogeneous Neumann data on the cut. Its even reflection solves a Dirichlet problem on the full polygon, with right-hand side the even extension of  $\Delta v_0$ . This belongs to  $H^{\sigma'}$  since  $\sigma' < 1/2$ . The Dirichlet shift theorem [4, Chapter 4] gives  $H^{2+\sigma'}$  regularity because  $1 + \sigma' < n/(n-2)$ , the first Dirichlet corner exponent. Adding back  $v_0$  proves the assertion on each half-polygon. This supplies the positive shift needed below; an  $H^2$  estimate alone would not suffice.

For odd  $n$ , the negative cut ends at an edge midpoint with zero Neumann data nearby; reflection gives a straight Dirichlet boundary there. Across the interior of that negative cut,  $W$  is harmonic and its first derivatives match. Its piecewise  $H^{\sigma'}$  second derivatives glue for  $\sigma' < 1/2$ , so the same regularity holds on each original fan sector even when the cut crosses a sector. The ray decomposition follows [3, Section 3]; the trace lifting above makes the endpoint compatibility explicit.  $\square$

**Proposition 13.2** (First material derivatives at the regular polygon). *For every fixed  $n \geq 3$ , the first vertex material derivatives are in  $H^{2+\sigma}$  on each coarse fan triangle for some  $\sigma > 0$  depending on  $n$ . Their gradients have a common value at the centre and vanish at the polygon vertices.*

*Proof.* Split (22) into its volume solution and three ray solutions. The volume load is piecewise  $H^\sigma$ , hence globally  $H^\sigma$  after choosing  $\sigma < 1/2$ ; multiplication by a sector indicator preserves  $H^\sigma$  in this range. The convex Dirichlet polygonal shift theorem gives  $H^{2+\sigma}$  for its solution, with a possibly smaller positive exponent. Lemma 13.1 handles the three ray solutions. Rotation and linear combination handle all vertices and all directions  $q$ .

The one-sided gradients are continuous to sector endpoints by Sobolev embedding. Across a ray, the tangential gradient agrees, because the function has one common trace. The normal-gradient jump equals the line density, up to the chosen orientation. That density vanishes at both endpoints, so the gradients agree at the centre and at each outer endpoint. At a polygon vertex the common gradient is orthogonal to both adjacent boundary tangents, hence is zero. At the centre the common gradient need not be zero; symmetry of the state does not imply the same symmetry for a single-vertex material derivative.  $\square$

## 14 The second derivatives: a necessary centre cancellation

The first-derivative proof is not automatically a proof for  $u_{qr}$ . Nevertheless it supplies the extra regularity needed to continue. On each sector write its weak equation as

$$a(u_{qr}, v) = \int_P d_{qr} v + \int_P G_{qr} \cdot \nabla v,$$

where, for this regularity calculation only,

$$\begin{aligned} d_{qr} &= (\operatorname{div} \theta_q)(\operatorname{div} \theta_r) - \operatorname{tr}(D\theta_q D\theta_r), \\ G_{qr} &= -\mathbb{A}_q \nabla u_r - \mathbb{A}_r \nabla u_q - \mathbb{A}_{qr} \nabla u. \end{aligned}$$

These two names display the volume and flux parts of one distribution; they do not introduce any new PDE solves. Sectorwise integration by parts produces  $d_{qr} - \operatorname{div} G_{qr} \in L^2$  and ray densities  $-[G_{qr}]_j \cdot \nu_j \in H^{1/2+\sigma}(S_j)$ . All outer endpoint values vanish by Proposition 13.2. The centre values may not vanish because  $\nabla u_q(0)$  and  $\nabla u_r(0)$  may be nonzero.

This matters. A constant load on a ray ending at an interior point can produce an  $r \log r$  corner term; its second derivatives are not square integrable near that endpoint. Splitting a compatible collection of rays into unrelated single-ray problems can therefore destroy the regularity one is trying to prove.

### 14.1 An explicit local lifting removes the centre values

Choose a smooth cutoff  $\chi$  supported in an interior disk around the centre, equal to one in a smaller disk. Subtract from  $u_{qr}$  the known piecewise smooth function

$$\chi(x)(\theta_q(x) \cdot \nabla u_r(0) + \theta_r(x) \cdot \nabla u_q(0)). \quad (23)$$

It belongs to  $H_0^1(P)$  and is  $H^2$  on each fan sector. Its purpose is local regularity analysis; it is not the numerical second lifting  $Z_{qr}^h$ . The centre gradients are well defined by the first-derivative proposition.

For any constant vector  $b$ , the following identity holds in distributions:

$$\operatorname{div}(-\mathbb{A}_q b) = \Delta(\theta_q \cdot b). \quad (24)$$

Indeed,  $-\mathbb{A}_q b = D\theta_q b + D\theta_q^T b - (\operatorname{div} \theta_q)b$ . The divergences of the first and last terms cancel by commutation of weak derivatives; the middle term is  $\nabla(\theta_q \cdot b)$ . Near the centre, where  $\chi = 1$ , identity (24) shows that (23) has exactly the ray-jump contributions of  $-\mathbb{A}_q \nabla u_r(0) - \mathbb{A}_r \nabla u_q(0)$ . Also  $\nabla u(0) = 0$ , so the  $\mathbb{A}_{qr}$  term contributes no centre value. Subtracting the lifting therefore makes every remaining ray density vanish at the centre.

Differentiating the cutoff introduces only piecewise smooth volume terms and ray terms supported away from the centre and the outer vertices. The resulting volume source is still  $L^2$ ; each ray density is still  $H^{1/2+\sigma}$  and now vanishes at *both* endpoints. The regular volume problem and the isolated ray problems can now be treated separately.

**Proposition 14.1** (Second material derivatives at the regular polygon). *The exact second material derivative  $u_{qr}$  belongs to  $H^2(T_j)$  for every fan triangle, every fixed regular polygon, and every pair of vertex directions  $q, r$ .*

*Proof.* After subtracting (23), solve the  $L^2$  volume problem with zero Dirichlet data. Its solution is globally  $H^2$  by convexity. For each remaining ray load apply Lemma 13.1. The sum of these solutions and the subtracted lifting satisfies the weak equation for  $u_{qr}$ , and uniqueness identifies it with  $u_{qr}$ . Every summand is  $H^2$  on each fan triangle. There are finitely many rays and finitely many vertex coordinates, so the argument applies to all mixed directions.  $\square$

## 14.2 Scope of the regularity and rate theorems

The argument proves the qualitative broken regularity of the state, first derivatives, and second derivatives *at*  $P_n$ , for fixed  $n$ . It also gives uniformity over bounded vertex directions there, by finite dimensionality and bilinearity. Constants may depend on  $n$ ; the corner exponent tends to one as  $n$  grows.

The manuscript's Theorem 6.3 now states the Hessian rate at this fixed regular polygon. Uniform broken regularity in a neighborhood of perturbed polygons is a stronger statement: those polygons no longer have the reflection symmetries used here. Such an extension would need stability of the transmission estimates under geometric perturbations, or a parameter-dependent corner expansion with uniform control.

For the Hessian rate *at the regular polygon*, the regularity just proved is sufficient. Broken interpolation and the differentiated Galerkin equations give  $\delta_0, \delta_q, \delta_r = O(h)$ . Moreover, subtracting the exact second-derivative equation from (13) gives

$$\left\| Z_{qr}^h - u_{qr} \right\|_a \leq C_q \delta_r + C_r \delta_q + C_{qr} \delta_0 = O(h).$$

Approximate the fixed function  $u_{qr}$  by its broken interpolant and use best approximation for  $Z_{qr}^h$  to obtain  $\eta_{qr} = O(h)$ . Equation (16) then gives  $D^2 J(P_n) - D^2 J_h(P_n) = O(h^2)$ . The energy identity and its first derivative also bound  $J - J_h$  and  $D(J - J_h)$  by  $O(h^2)$ , so differentiating  $A^{-2}$  gives the same rate for the normalized Hessian. This is a statement about exact Galerkin quantities, not yet about the computed radius in (18).

### Exercises and proof checkpoints

**Exercise 14.2.** For  $c = 2$ , simplify the three ray terms in (22) using  $\nabla u(ra_j) = a_j \partial_r u(r, 0)$ .

*Answer.* The  $S_0$  term is zero. The terms on  $S_1, S_{-1}$  are respectively  $-\partial_r u \delta_{S_1}$  and  $+\partial_r u \delta_{S_{-1}}$ , with the traces understood on their rays. For  $c = 1$  the weights are  $2 \cot \vartheta, -\cot \vartheta, -\cot \vartheta$ .

**Exercise 14.3.** Why is pointwise zero at an endpoint not an argument for a general  $H^{1/2}$  function?

*Answer.* Endpoint evaluation is not a continuous trace at the borderline exponent. The positive extra regularity above, or an explicit weighted endpoint condition, must be used.

**Exercise 14.4.** Which extra facts are needed to conclude that the computed  $R_F$  is  $O(h^2)$ ?

*Answer.* The four chosen flux mismatches and the transferred algebraic bounds must have compatible orders, so that  $B_J, E_{\text{alg}}$ , and the normalization remainder are  $O(h^2)$ . A fixed solver tolerance or fixed precision can eventually prevent that asymptotic behavior even when the exact Galerkin Hessian converges quadratically.

## Part III

# Reading and running the FLINT certificate

## 15 What the executable has to prove

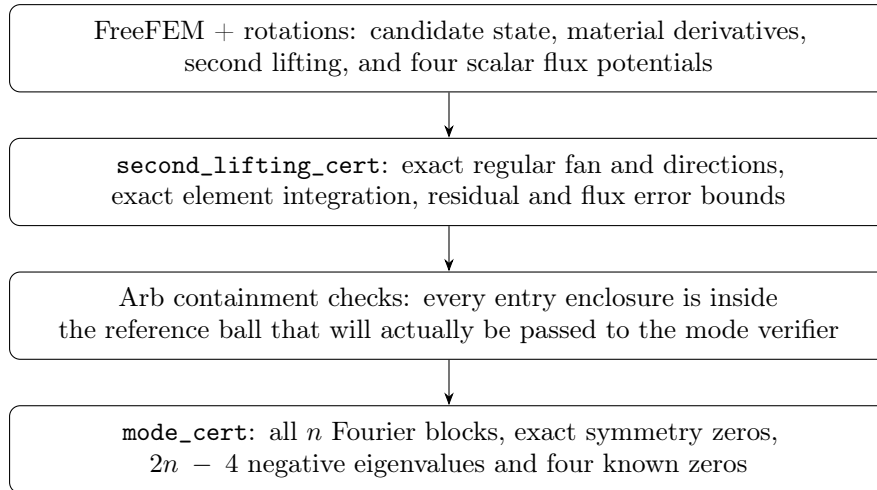
**Aim.** Read the maintained implementation as a sequence of mathematical assertions. A suggested 90-minute session devotes 15 minutes to the proof contract, 35 to the entry verifier, 20 to the Fourier verifier, and 20 to reproduction and results. The function descriptions also serve as a reference for a longer practical session with the source open.

For the circumradius-one regular polygon, let

$$-\Delta u = 1, \quad u|_{\partial P_n} = 0, \quad J = \frac{1}{2} \int_{P_n} u, \quad \mathcal{F} = J/|P_n|^2.$$

The conclusion sought is that the vertex Hessian of  $\mathcal{F}$  has exactly  $2n - 4$  strictly negative eigenvalues and the four zero eigenvalues imposed by two translations, rotation, and dilation. Lecture 1 explains why this implies strict local maximality modulo similarities. Lecture 2 supplies the error bounds for the individual Hessian entries.

There are two different finite computations. The first encloses exact PDE-dependent Hessian entries. The second converts those enclosures into eigenvalue signs. The following diagram is also the order in which to read the implementation.



FreeFEM and NumPy propose finite element functions. Their linear solves, floating geometry, and FFTs need not be exact: the C verifier checks the geometry and directions, then bounds the errors of the supplied functions. The mathematical identities, the implementation of the two verifiers, and FLINT/Arb's arithmetic remain part of the trusted argument. This is a computer-assisted proof, rather than a proof object checked by a formal proof assistant.

The current source root is `MaxTorsionValidation/`. The accompanying `source_index.md` gives function locations, and `sources.json` records the hashes of the files used for these notes. Descriptions below refer to that snapshot, so a reader can distinguish it from sources saved inside an older computation archive.

## 16 The finite element data entering one entry check

Write  $\vartheta = 2\pi/n$ . Subdivide each coarse fan triangle

$$T_j = \text{conv}\{0, (\cos j\vartheta, \sin j\vartheta), (\cos((j+1)\vartheta), \sin((j+1)\vartheta))\}$$

uniformly with  $m$  segments on each side. Then

$$\#\text{triangles} = nm^2, \quad \#\text{nodes} = 1 + \frac{nm(m+1)}{2}, \quad \#\text{boundary nodes} = nm.$$

Every fine triangle is contained in one coarse sector. Thus the gradients of the fan extensions and the pullback coefficients are constant on each fine triangle. This fitted structure is used both in the regularity-based rate discussion and in the exact coefficient reconstruction.

An export starts with

```
TORSION_SECOND_V2 n m k qa qp ra rp nv nt
```

Here  $k$  is the Fourier index; **qa**, **ra** select radial (0) or tangential (1) directions, and **qp**, **rp** select cosine (0) or sine (1). The final two integers are the node and triangle counts. Next come the node coordinates, the three node indices of each triangle, and twelve nodal arrays, in this order:

```
qx qy rx ry u uq ur zh psi0 psiq psir psiz
```

The first four arrays are components of the fan extensions  $\theta_q, \theta_r$ . The next four specify conforming candidates for the state, two first material derivatives, and the second lifting. The final four specify continuous piecewise affine scalar potentials whose curls are used in the equilibrated fluxes. An array named **u** is therefore a computed candidate, not the exact torsion solution.

For a nonzero index other than the even Nyquist index, the normalized real directions are

$$\begin{aligned} r_{k,j}^c &= \sqrt{2/n} \cos(kj\vartheta)(\cos j\vartheta, \sin j\vartheta), & \text{radial, cosine,} \\ t_{k,j}^c &= \sqrt{2/n} \cos(kj\vartheta)(-\sin j\vartheta, \cos j\vartheta), & \text{tangential, cosine,} \\ t_{k,j}^s &= \sqrt{2/n} \sin(kj\vartheta)(-\sin j\vartheta, \cos j\vartheta), & \text{tangential, sine.} \end{aligned}$$

Using the Fourier convention of Lecture 1, the four requested entries are the following real Hessian pairings:

| Entry name   | $q$     | $r$     | Exact value of $D^2\mathcal{F}[q, r]$ |
|--------------|---------|---------|---------------------------------------|
| <b>rr</b>    | $r_k^c$ | $r_k^c$ | $\alpha_k$                            |
| <b>tt</b>    | $t_k^c$ | $t_k^c$ | $\beta_k$                             |
| <b>rt_re</b> | $r_k^c$ | $t_k^c$ | $\text{Re}(B_k)_{12} = 0$             |
| <b>rt_im</b> | $r_k^c$ | $t_k^s$ | $\text{Im}(B_k)_{12} = \gamma_k$      |

For  $n$  even and  $k = n/2$ , replace  $\sqrt{2/n}$  by  $1/\sqrt{n}$  and omit the zero sine direction. Only the two diagonal pairings are then needed.

## 17 Arb arithmetic: enclosures rather than rounded answers

An Arb real ball represents a midpoint with a rigorous absolute error radius. Arithmetic propagates enclosures; increasing the precision reduces rounding error but does not remove finite element error. The maintained driver uses 192 bits in both verification stages. The relevant documented operations are: **arb\_set\_str** for decimal input, **arb\_add\_error** for enlarging a radius, **arb\_contains** for containment, and **arb\_is\_negative** for strict negativity of the whole ball [10].

In particular, a negative midpoint is insufficient. If an enclosure is  $[-3 \cdot 10^{-5}, 10^{-6}]$ , its sign is unresolved. Similarly, printed digits are not the internal ball. The driver may use printed output to propose a convenient radius, but the entry verifier must subsequently prove that the exact reference ball contains its internal enclosure.

The code uses **arb\_sqrtpos** only for expressions mathematically known to be nonnegative, such as sums of squares. Intersecting their enclosure with the nonnegative half-line is justified by that identity; it would not justify taking a square root of an arbitrary uncertain input. The input readers reject nonfinite numbers before they are used in the certificate.

## 18 The entry verifier, function by function

Open `MaxTorsionValidation/flint/second_lifting_cert.c`. The program does not solve a large interval linear system. It integrates the supplied piecewise affine fields and evaluates the error inequalities of Lecture 2. Its functions fall into four short groups, followed by the main calculation.

### 18.1 Reading a finite conforming mesh

**die.** Print an explanation and exit with status 2. A failed input assertion is an error, not a certificate with a large radius.

**cmp\_edge and cmp\_tri\_key.** Compare sorted vertex pairs and triples lexicographically. Sorting makes edge multiplicities and duplicate triangles detectable independently of their order in the exported file.

**dsu\_root and dsu\_join.** Maintain connected components using a disjoint-set data structure, with path compression. They supply the connectivity part of the mesh check.

**certify\_mesh\_topology.** Check valid and distinct triangle vertices, reject duplicate triangles, require all nodes to be used and connected, count edges with one or two incident triangles, and check the boundary degrees, boundary count, and Euler relation. In strict mode the node and triangle counts must equal the regular-fan formulas above. The function returns the boundary mask. These combinatorial conditions are necessary; the later equality with the complete canonical triangulation establishes the particular mesh required by the proof.

**read\_ball and read\_field.** Read a finite decimal as an Arb ball and optionally enlarge it by a specified radius; repeat for a nodal array. Geometry and direction candidates are initially given radii  $10^{-13}$ . PDE and potential coefficients are read with no additional modelling radius: the decimal coefficients define the candidate finite element function, with conversion rounding enclosed by Arb. Truncation and nonfinite data are rejected.

**assemble\_triangle\_geometry.** Form the signed determinant of the two triangle edge vectors, prove it does not contain zero, and compute the area and the gradients of the three barycentric basis functions. The signed determinant is used for gradients; its absolute value is used for area. Geometry is assembled again after the exact regular coordinates have replaced the candidate coordinates.

### 18.2 Proving the exact regular fan and the directions

**canonical\_gid.** Give each lattice node of the regular fan a single integer index. The centre has one index; nodes on shared rays have the same index when described from either neighboring sector. This is how the program avoids counting a shared ray twice.

**canonical\_tri\_key.** Sort the three canonical node indices of a triangle. This removes an irrelevant orientation or input-order choice when comparing triangulations.

**validate\_regular\_inputs.** Reconstruct the exact node set

$$x = \frac{s}{m}(\cos j\vartheta, \sin j\vartheta) + \frac{t}{m}(\cos((j+1)\vartheta), \sin((j+1)\vartheta)), \quad s, t \geq 0, \quad s + t \leq m.$$

The integer pair  $(s, t)$  is a lattice address, not a new angle notation. Floating calculations nominate an address; Arb then proves that the candidate coordinate balls contain the trigonometric coordinates of that address. The assignment must be a bijection. The boundary mask must agree with the canonical boundary, and the sorted list of triangles must equal the full canonical list, not merely have the correct cardinality.

The function also reconstructs the normalized Fourier directions at each node and proves their containment in the exported direction balls. It finally replaces geometry and direction arrays by these analytic enclosures. Each triangle has exact area  $\sin \vartheta / (2m^2)$  and the total area is  $n \sin \vartheta / 2$ . Consequently the  $10^{-13}$  input radii serve to identify the intended exact data; they do not certify an arbitrary perturbation of the polygon or enter as a shape uncertainty.

### 18.3 Exact element integration and coefficient bounds

**gradient.** For a nodal scalar field  $v$ , compute  $\nabla v|_T = \sum_{i=0}^2 v_i \nabla \lambda_i$ . It is constant on a triangle, so no numerical quadrature is required.

**affine\_integral.** Use  $\int_T v = |T|(v_0 + v_1 + v_2)/3$ .

**add\_affine\_square.** Add the exact integral

$$\int_T v^2 = \frac{|T|}{6} (v_0^2 + v_1^2 + v_2^2 + v_0 v_1 + v_0 v_2 + v_1 v_2). \quad (25)$$

This is the usual consistent  $P_1$  mass matrix. As a quick check,  $v_0 = v_1 = v_2 = 1$  gives  $|T|$ . Applied separately to the two components of an affine flux mismatch, it gives its squared  $L^2$  norm exactly, with only interval arithmetic rounding to enclose.

**symmetric\_norm.** For a real symmetric matrix with entries  $x, y, z$ , compute its operator norm from

$$\left\| \begin{pmatrix} x & y \\ y & z \end{pmatrix} \right\|_2 = \frac{|x + z| + \sqrt{(x - z)^2 + 4y^2}}{2}.$$

Taking a maximum over triangles supplies the constants  $C_q = \|\mathbb{A}_q\|_{L^\infty}$ ,  $C_r$ , and  $C_{qr} = \|\mathbb{A}_{qr}\|_{L^\infty}$  used in Lecture 2.

**coefficients.** Differentiate the planar pullback coefficient

$$\mathbb{A}(s, t) = \det(\text{Id} + sD\theta_q + tD\theta_r)(\text{Id} + sD\theta_q + tD\theta_r)^{-1}(\text{Id} + sD\theta_q + tD\theta_r)^{-T}.$$

The function returns  $\mathbb{A}_q, \mathbb{A}_r, \mathbb{A}_{qr}$  and the mixed determinant derivative

$$(\text{div } \theta_q)(\text{div } \theta_r) - \text{tr}(D\theta_q D\theta_r).$$

The first matrix derivative is  $\mathbb{A}_q = (\text{div } \theta_q) \text{Id} - D\theta_q - D\theta_q^T$ . To read the second derivative without an extra chain of aliases, expand it as

$$\begin{aligned} \mathbb{A}_{qr} = & ((\text{div } \theta_q)(\text{div } \theta_r) - \text{tr}(D\theta_q D\theta_r)) \text{Id} \\ & - (\text{div } \theta_q)(D\theta_r + D\theta_r^T) - (\text{div } \theta_r)(D\theta_q + D\theta_q^T) \\ & + D\theta_q D\theta_r + D\theta_r D\theta_q + D\theta_q^T D\theta_r^T + D\theta_r^T D\theta_q^T \\ & + D\theta_q D\theta_r^T + D\theta_r D\theta_q^T. \end{aligned}$$

The source stores each symmetric matrix in three entries. Its temporary four-entry arrays named **Q** and **R** are simply the row-major Jacobians  $D\theta_q, D\theta_r$ ; no additional mathematical quantities are needed to understand them.

**state\_error.** Compute

$$\Phi_0 = \| -x/2 + \text{curl } \tilde{\psi}_0 - \nabla \tilde{u} \|_{L^2(P_n)}, \quad \text{curl } \psi = (\partial_y \psi, -\partial_x \psi).$$

The flux has divergence  $-1$ . Continuity of the scalar potential ensures the required normal continuity of its curl. Thus  $\Phi_0$  bounds the state energy error even if the auxiliary fit for  $\tilde{\psi}_0$  is inexact.

## 18.4 Residuals, reference values, and the main calculation

**interior\_residual\_norm.** Take the Euclidean norm of a residual array on the interior nodes only. The basis of  $V_h \subset H_0^1(P_n)$  has no boundary degrees of freedom. Including the boundary rows of the unconstrained stiffness matrix would therefore compute the wrong algebraic residual for this problem.

**read\_mode\_reference.** Read the requested component of a `MODE_DATA` record: `rr`, `tt`, `rt_re`, or `rt_im`. These correspond respectively to the two diagonal entries and the real and imaginary parts of the upper-right entry. The entry pass checks the requested index and its conjugate; the final mode pass checks uniqueness and completeness of all Fourier records.

**main: assertions before the triangle loop.** Require exactly one of `-check-regular-inputs` and `-diagnostic-nonstrict`. Reference certification requires the strict choice, the reference file, the entry name, and the proposed maximum radius. Validate dimensions, metadata, and radii; read the export; reject trailing data. Set the four Dirichlet candidate fields to exact zero on the certified boundary. The scalar flux potentials have no Dirichlet condition. Then validate and reconstruct the exact regular inputs.

**main: the triangle loop.** Accumulate the area,  $\int \tilde{u}$ , the second area variation, the raw Hessian centre, the three coefficient norm bounds, the four residual vectors, the node patch areas, and the three remaining squared flux mismatches. The state mismatch was already evaluated by **state\_error**. Each calculation uses constant gradients and (25); there is no quadrature remainder to estimate.

The mathematical residual convention is  $\rho = \text{right side} - \text{matrix} \times \text{candidate}$ . The C arrays `res0`, `resq`, `resr`, `resz` store its *negative*. Norms do not change, but the sign matters when correcting the centre.

**main: algebraic errors.** The polygon lies in  $(-1, 1)^2$ , so domain monotonicity gives  $\lambda_1(P_n) \geq \pi^2/2$ . The local mass matrix gives, for the interior-node stiffness matrix  $K_h$ ,

$$K_h \succeq \alpha \text{Id}, \quad \alpha = \frac{\pi^2}{24} \min_{i \text{ interior}} \sum_{T \ni i} |T| > 0.$$

Thus the dual norm of a residual restricted to  $V_h$  is bounded by its Euclidean coefficient norm divided by  $\sqrt{\alpha}$ . The first-derivative algebraic bounds also contain the state error multiplied by  $C_q$  or  $C_r$ , because the right sides depend on the state candidate.

The corrected centres are those derived in Lecture 2:

$$\tilde{C}_{\text{corr}} = \tilde{C} + \rho_r(\tilde{u}_q) + \rho_0(\tilde{z}), \quad \tilde{J}_{\text{corr}} = \int_P \tilde{u} - \frac{1}{2} \int_P |\nabla \tilde{u}|^2.$$

In the C loop, adding  $\rho_r(\tilde{u}_q)$  means subtracting `resr` dotted with `uq`. The remaining algebraic error in the Hessian centre is quadratic in the relevant energy errors.

**main: the final entry enclosure.** Let  $\widehat{\delta}_0, \widehat{\delta}_q, \widehat{\delta}_r$  and  $\widehat{\eta}_{qr}$  denote the computed upper bounds for the state, first-derivative, and second-lifting Galerkin errors, and let  $E_{\text{alg}}$  bound the corrected discrete-centre error. These are the quantities of Lecture 2, not new error definitions. For the nonzero Fourier pairings the first area variations vanish. The program evaluates

$$\begin{aligned}\widetilde{F}_{qr} &= \frac{\widetilde{C}_{\text{corr}}}{A^2} - \frac{2\widetilde{J}_{\text{corr}}A_{qr}}{A^3}, \\ R_F &= \frac{\widehat{\delta}_q\widehat{\delta}_r + \frac{1}{2}C_{qr}\widehat{\delta}_0^2 + \widehat{\delta}_0\widehat{\eta}_{qr} + E_{\text{alg}}}{A^2} + \frac{|A_{qr}|}{A^3}(\widehat{\delta}_0^2 + \epsilon_0^2).\end{aligned}$$

Here  $A = |P_n|$ ,  $A_{qr} = D^2A[q, r]$ , and  $\epsilon_0$  bounds the state algebraic energy error. It enlarges the ball for  $\widetilde{F}_{qr}$  by  $R_F$ . Its output separates `F_hessian_continuous_error<=` from `F_hessian_algebraic_error<=`, making the source of a large radius visible.

**main: the handoff assertion.** If a reference is supplied, reconstruct its ball using the proposed common mode radius and prove that it contains the complete entry enclosure. Repeat at the conjugate mode  $n - k$ ; negate the imaginary entry there. Only successful containment prints

```
ENTRY_CERTIFIED k=... conjugate_k=... entry=... max_radius=...
```

The driver checks that all four metadata fields match its request. This second pass closes the gap between a rigorously computed internal ball and the finite decimal data actually sent to the eigenvalue verifier.

## 19 The Fourier verifier, function by function

Open `MaxTorsionValidation/flint/mode_cert.c`. Its input consists of

```
MODE_DATA k a d Re(b) Im(b)
```

representing  $\begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$ , and a radius file with one `MODE_RADIUS k radius` line for every index. The letters in this file format mean the two diagonal entries and the complex upper-right entry; in Lecture 1 these are  $\alpha_k, \beta_k$ , and  $i\gamma_k$ .

**usage.** Describe the accepted options and exit with status 2. A global radius and a per-mode radius file cannot be used together.

**print\_ball.** Print an enclosure for inspection. The displayed precision does not replace the precision used in the sign predicates.

**main: input validation.** Read finite real components and nonnegative radii. With the expected number of symbols specified, require exactly one mode and one radius for every  $k = 0, \dots, n - 1$ . Reject missing, duplicate, or out-of-range indices. These checks prevent a repeated negative mode from standing in for an untested one.

**main: analytic symmetry information.** With `-exact-regular-similarities`, replace  $B_0$  by the exact zero matrix. At  $k = 1, n - 1$ , use the exact translation eigenvalue zero and the trace for the other eigenvalue. At the even Nyquist index, set the off-diagonal entries to zero. These are analytic identities from Lecture 1; they are applicable because the first stage proved that the data correspond to the regular polygon and its normalized Fourier directions.

**main: eigenvalues and sign counts.** For the remaining blocks evaluate

$$\lambda_{\pm} = \frac{a + d \pm \sqrt{(a - d)^2 + 4(\operatorname{Re} b)^2 + 4(\operatorname{Im} b)^2}}{2}.$$

Count an eigenvalue as negative only when its entire Arb ball is strictly negative. The internal word **unresolved** includes the four exact zeros. In a successful regular-polygon run these are its only members. An interval proved positive is not counted as unresolved; it also cannot satisfy the required negative count.

The full driver requires all three counts

$$\text{symbols} = n, \quad \text{negative} = 2n - 4, \quad \text{unresolved} = 4.$$

The program exits 0 when the requested counts hold, 3 when they fail, and 2 on an input error. Its printed **MODE\_CERTIFIED** marker means the *requested* counts were satisfied. A standalone invocation that requests weaker counts does not prove local maximality. Nor does this program prove the PDE error radii: those must come from the entry checks.

## 20 How the rotation-based driver obtains the candidates

The optimization concerns the first derivatives. At vertex 0, the local radial and tangential directions are (1,0) and (0,1). Solve the torsion problem and these two material problems. Rotational symmetry gives the other local vertex derivatives by composing these scalar functions with the inverse rotation. Weighted sums give the Fourier derivatives. Thus only three primal state/first-derivative solves are needed per polygon. The present error estimator still solves a second lifting for each real Hessian pairing, and it computes auxiliary scalar flux fits. Counting those as well would give more than three linear solves.

### 20.1 The FreeFEM problems

In `freefem/second_variation_lifting_rotations.edp`, the named problem **state** is the torsion Galerkin equation, **firstQ** and **firstR** are the two differentiated Galerkin equations, and **lifting** is the discrete second lifting of Lecture 2. The `-make-basis 1` branch solves the first three once at vertex 0. The `-cached-fields` branch reads the state and the requested first derivatives from the cache and solves only **lifting**.

The named problem **fitCurl** minimizes a regularized flux mismatch over continuous piecewise affine scalar potentials. The added term is  $10^{-12} \int_P \psi_h^2 / 2$  in the minimized functional: it makes the fit coercive and fixes the additive constant, while also slightly changing the nonconstant part of the minimizer. Three such fits accompany the initial basis; one more accompanies each second lifting. The argument for equilibrium uses the form of the flux, not successful convergence of this optimization. The solver tolerance affects sharpness and speed, while the C verifier evaluates the actual supplied potential.

The separate `freefem/torsion_hessian_rotations.edp` assembles a floating Hessian from the same three-solve principle. It is useful for inspection and comparisons. Its named problems **torsion**, **materialX**, and **materialY** solve the state and the two coordinate material equations at vertex zero. The driver option `-entry-centers` avoids needing this additional floating Hessian by taking reference centres directly from the entry estimates.

### 20.2 rotation\_cache.py

**prepare.** Read the basis export, check dimensions and finiteness, identify lattice addresses, construct the node permutation for rotation, and check its cycles and coordinate consistency. Shared-ray addresses have one canonical key. Apply the permutation to the two vertex-zero

scalar solutions and their flux potentials, then form Fourier combinations with NumPy’s FFT. For NumPy’s negative-exponent convention, cosine weights come from the real part and positive sine weights from the negative imaginary part. The normalization is  $\sqrt{n}/2$ , or  $\sqrt{n}$  for the even Nyquist mode, when dividing an unnormalized Fourier sum.

The nested function `direction` selects the radial/tangential and cosine/sine combination for one requested field. The cache is only a candidate-generation device: exact geometric and directional checks, and residual bounds for the resulting fields, are repeated in C.

### 20.3 `certificate_build.py`

**build\_verifiers.** Compile `second_lifting_cert` and `mode_cert` from the source copies saved inside the new archive, using a forced build. Return their private executable paths to the driver and record their hashes. A small diagnostic records the compiler and FLINT header/runtime versions. Failure of that diagnostic is recorded as unavailable metadata; the two verifier builds themselves must succeed. The run therefore does not silently use a pre-existing executable from a different source version.

### 20.4 `certify_with_rotations.py`

**run.** Execute a subprocess, save its output in a designated log, measure elapsed time, and raise an error for a nonzero exit status. The final mode step is handled separately so that an inconclusive certificate can be recorded normally rather than mistaken for a crashed program.

**ball.** Read the printed midpoint and radius for a named output using decimal arithmetic. Its result is used to propose a radius. It has no independent authority to prove containment.

**main: setup and candidate phase.** Parse the range of  $n$ ,  $m$ , worker count, CPU affinity limit, flux-fit tolerance, and a new archive path. Limit numerical-library threads to one per process, set the chosen affinity, and run with reduced scheduling priority. Save a source snapshot and manifest, build the private verifiers, and create the three-solve rotation cache for each polygon.

The nested function `evaluate` generates one second-lifting export, checks its header against the requested pairing, and runs the entry verifier in strict mode. From the displayed centre and error ball it proposes a slightly enlarged decimal radius covering both  $k$  and  $n - k$ . Four pairings are used for each representative nonzero mode, except the even Nyquist mode, which needs two.

**main: reference and verification phase.** Once all centres exist, write `modes.txt` once and retain it unchanged during verification. Take the maximum proposed entry radius for each mode and its conjugate. The nested function `verify` reruns the C entry verifier with this exact reference file and radius, checks the complete `ENTRY_CERTIFIED` marker, and compresses the checked export. After all entries pass, run `mode_cert` at 192 bits with the required  $n, 2n - 4, 4$  counts. Save `RESULT.json`, append `results.jsonl`, and stop on an inconclusive polygon if `-stop-on-failure` is set. Finally save archive hashes and a `FINISHED` marker.

**Reading a run’s status.** The Python driver can exit normally after recording `NOT_CERTIFIED`; this is an expected numerical outcome. `FINISHED` means that the requested sweep or its stopping rule has finished. To claim success for a particular  $n$ , inspect its `RESULT.json`: it must say `status: CERTIFIED` and `mode_exit_code: 0`, with the corresponding entry logs and final counts present. A shell exit code alone is insufficient.

## 21 Other validation utilities and their precise roles

The two C programs above are the maintained complete certification path. The repository also contains smaller verifiers and diagnostic utilities. Their output should be read according to the narrower assertion each one actually establishes.

### 21.1 `residual_cert.c`

`read_arb` parses a scalar. Its `main` reads a dense matrix, right side, candidate vector, and a supplied positive lower bound  $\alpha$ ; it evaluates the residual and the resulting algebraic error bound. The assumption that the matrix is symmetric positive definite and that  $\alpha \leq \lambda_{\min}$  is supplied by the caller. Positivity of the number  $\alpha$  alone does not prove that assumption. This utility does not perform the regular-fan and boundary checks of the main verifier.

### 21.2 `second_variation_cert.c`

`die` reports invalid input; `set_nonnegative` parses and checks nonnegative bounds. The `main` function reads complete records of two direction indices and five bounds, and evaluates

$$\delta_q \delta_r + \frac{1}{2} C_{qr} \delta_0^2 + \delta_0 \eta_{qr}.$$

It checks the arithmetic combination, assuming that the five supplied bounds have already been proved. It does not establish their PDE meaning.

### 21.3 `majorant_cert.c`

| Function                                        | Role                                                                                    |
|-------------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>die</code>                                | Reject an invalid export.                                                               |
| <code>read_arb</code> , <code>read_field</code> | Read scalar balls and arrays.                                                           |
| <code>grad_field</code>                         | Compute a $P_1$ gradient on a triangle.                                                 |
| <code>add_affine_square</code>                  | Integrate a squared affine mismatch.                                                    |
| <code>energy_norm</code>                        | Integrate a field's squared gradient and take its norm.                                 |
| <code>state_majorant</code>                     | Bound the state error by its equilibrated flux mismatch.                                |
| <code>material_majorant</code>                  | Bound a first material error, including propagation of the state error.                 |
| <code>print_ball</code>                         | Display an enclosure.                                                                   |
| <code>main</code>                               | Read candidate data and combine direct state/material majorants into diagnostic bounds. |

This program is useful for understanding individual majorants and direct Gram estimates. It does not establish the complete canonical-fan input contract used by `second_lifting_cert`; its output alone is not the maintained local-maximality certificate.

### 21.4 Drivers, summaries, and checks

`freefem/test_extended_certificates.py` follows the same two-stage certificate structure with per-entry candidate solves instead of a shared rotation cache. Its `run`, `ball`, nested `evaluate` and `verify`, and `main` have the corresponding roles described above. The original candidate-generating FreeFEM files remain available for comparisons.

In `freefem/summarize_extended_certificates.py`, `audit` reads an archive's metadata, source hashes, entry markers, radii, and mode output; `main` assembles the requested result

tables. Table upper endpoints are rounded outward using decimal arithmetic. This is an audit of records, not a fresh replay of every finite element export.

The input-rejection tests in `flint/test_certificate_inputs.py` and the rotation comparisons in `freefem/test_torsion_hessian_rotations.py` exercise implementation properties. Neither a successful comparison with floating output nor a successful test suite substitutes for the entry enclosures and sign counts for a particular polygon.

## 22 Running a certificate yourself

The commands below are run from the repository root. They require FreeFEM, a C compiler with FLINT/Arb development files, Python, and NumPy; the maintained installation instructions are in `MaxTorsionValidation/README.md`. The archive directory must not already exist, so each run preserves its own evidence.

For a small first run:

```
python3 MaxTorsionValidation/freefem/certify_with_rotations.py \
  5 5 --m 32 --entry-centers --jobs 6 --cpu-cores 6 \
  --stop-on-failure --archive /tmp/torsion_n5_m32_course
```

For the largest successfully archived case:

```
python3 MaxTorsionValidation/freefem/certify_with_rotations.py \
  25 25 --m 128 --entry-centers --jobs 6 --cpu-cores 6 \
  --flux-eps 1e-13 --stop-on-failure \
  --archive /tmp/torsion_n25_m128_course
```

To test a range, replace the two positional integers by its first and last  $n$ . The stopping option halts after the first failure to certify. For this project's six-physical-core machine the shown settings use six workers. More generally, `-cpu-cores` selects from OS CPU affinity indices; it does not discover physical-core topology on another machine.

Inspect a completed run with, for example,

```
cat /tmp/torsion_n25_m128_course/n25/RESULT.json
tail -n 2 /tmp/torsion_n25_m128_course/n25/mode_cert.log
```

The expected final counts for  $n = 25$  are 25 symbols, 46 negative eigenvalues, and four exact similarity zeros. The archive also contains the reference centres, per-mode radii, compressed entry exports, entry logs, source snapshot, build metadata, and hashes.

An independent replay of a single entry uses the archived verifier and the decompressed export, with the same `-prec 192`, geometry and direction radii, strict input option, reference file, entry name, and `-max-radius` as its original command. Running only the final `mode_cert` is a useful replay of the small spectral calculation, but it does not replay the entry PDE bounds.

## 23 What the archived computations establish

The following tables transcribe the manuscript and the archived table data. No new certification sweep was run to write these notes. All eigenvalue values refer to  $D^2(J/A^2)$  in the normalized vertex coordinates of Lecture 1. The listed upper bound is the largest certified upper endpoint among eigenvalues other than the four similarity zeros. Negative upper bounds are rounded toward  $+\infty$ , so the displayed values remain safe upper bounds.

| $n$ | $m$ | Negative / required | Upper bound           |
|-----|-----|---------------------|-----------------------|
| 3   | 32  | 2/2                 | $-1.00 \cdot 10^{-2}$ |
| 4   | 32  | 4/4                 | $-6.20 \cdot 10^{-3}$ |
| 5   | 32  | 6/6                 | $-3.10 \cdot 10^{-3}$ |
| 6   | 32  | 8/8                 | $-1.70 \cdot 10^{-3}$ |
| 7   | 32  | 10/10               | $-9.80 \cdot 10^{-4}$ |
| 8   | 32  | 12/12               | $-5.90 \cdot 10^{-4}$ |
| 9   | 32  | 14/14               | $-3.60 \cdot 10^{-4}$ |
| 10  | 32  | 16/16               | $-2.20 \cdot 10^{-4}$ |

The  $n = 3, 4$  cases are also known analytically. The computations extend the local-maximality result through  $n = 25$ . Four similarity zeros accompany every successful row; they are known exactly from invariance, not inferred from small decimal numbers.

### The $m = 64$ computations

| $n$ | Triangles | Negative / required | Upper bound           | Outcome      |
|-----|-----------|---------------------|-----------------------|--------------|
| 11  | 45056     | 18 / 18             | $-2.11 \cdot 10^{-4}$ | certified    |
| 12  | 49152     | 20 / 20             | $-1.50 \cdot 10^{-4}$ | certified    |
| 13  | 53248     | 22 / 22             | $-1.08 \cdot 10^{-4}$ | certified    |
| 14  | 57344     | 24 / 24             | $-7.99 \cdot 10^{-5}$ | certified    |
| 15  | 61440     | 26 / 26             | $-5.97 \cdot 10^{-5}$ | certified    |
| 16  | 65536     | 28 / 28             | $-4.51 \cdot 10^{-5}$ | certified    |
| 17  | 69632     | 30 / 30             | $-3.44 \cdot 10^{-5}$ | certified    |
| 18  | 73728     | 32 / 32             | $-2.64 \cdot 10^{-5}$ | certified    |
| 19  | 77824     | 30 / 34             | $1.35 \cdot 10^{-5}$  | inconclusive |
| 20  | 81920     | 28 / 36             | $1.69 \cdot 10^{-4}$  | inconclusive |

### The $m = 128$ computations

| $n$ | Triangles | Negative / required | Upper bound           | Minutes | Outcome   |
|-----|-----------|---------------------|-----------------------|---------|-----------|
| 19  | 311296    | 34 / 34             | $-2.73 \cdot 10^{-5}$ | 15.7    | certified |
| 20  | 327680    | 36 / 36             | $-2.22 \cdot 10^{-5}$ | 18.0    | certified |
| 21  | 344064    | 38 / 38             | $-1.81 \cdot 10^{-5}$ | 20.8    | certified |
| 22  | 360448    | 40 / 40             | $-1.49 \cdot 10^{-5}$ | 13.5    | certified |
| 23  | 376832    | 42 / 42             | $-1.24 \cdot 10^{-5}$ | 15.8    | certified |
| 24  | 393216    | 44 / 44             | $-1.03 \cdot 10^{-5}$ | 19.4    | certified |
| 25  | 409600    | 46 / 46             | $-8.71 \cdot 10^{-6}$ | 17.7    | certified |

At  $m = 64$ , the archived checks succeed through  $n = 18$  and are inconclusive for  $n = 19, 20$ . A positive upper endpoint in those rows does not prove a positive eigenvalue. It means the enclosure is too wide to establish all the desired negative signs. Refinement to  $m = 128$  succeeds for every tested  $n = 19, \dots, 25$ . These computations neither establish a maximal possible  $n$  at either mesh size nor give a conclusion beyond  $n = 25$ .

### 23.1 Time and the meaning of the three-solve optimization

The recorded  $m = 128$  run for  $n = 25$  took 1059.4 seconds, about 17.7 minutes, on the six-physical-core Intel Core i7-9750H machine (2.60 GHz). It has 409600 triangles, 206401 nodes, and 48 real entry checks. The full optimized  $n = 22, \dots, 25$  records total about 66.4 minutes; allowing roughly 70 minutes is a useful estimate on that machine. The  $n = 24$  record includes a handoff interval, and the earlier  $n = 19, 20, 21$  runs used a different allocation/workflow, so the timings do not form a controlled scaling benchmark.

The three initial primal solves do not make the whole proof constant cost in  $n$ : there are more modes, more mesh triangles, one second lifting and one auxiliary flux fit per real pairing, and two Arb entry passes. This explains why the optimization is substantial while certification still takes minutes.

## 23.2 Where the evidence is stored

The main archive groups are under `MaxTorsionValidation/results/`:

- `review_2026_09_08_certificate_archive`: the  $m = 32$  chain.
- `extended_m128_fast_n19_onward`: the  $m = 128$ ,  $n = 19, 20, 21$  records.
- `extended_m128_rotations_n22_n30`: the completed  $n = 22, 23, 24$  records; the folder name records an initial range, not results through 30.
- `extended_m128_rotations_n25`: the separate  $n = 25$  certificate.
- `extended_m64_table_data.json` and `extended_m128_table_data.json`: the audited table rows and their per-row log references.

The packaging rename from `code/` to `MaxTorsionValidation/` preserved historical archives byte for byte. Consequently an old JSON record can still contain a `code/results/...` path; resolve that prefix to the current source root when locating the file. The packaging provenance explains this explicitly.

The recorded validity review checked 672 entry-log records, replayed 25 final mode checks, and performed four complete entry replays. Its archive hash audit covered 2701 files and recorded one historical checksum mismatch for the compressed  $n = 21$ ,  $k = 6$ , radial-radial export. That entry was freshly recertified at the archived radius; the original checksum record was retained and the replay documented in `validity_review_20260909/n21-replay-provenance.json`. This is the precise scope of that review: it was not a fresh recomputation of all 672 entry exports. Hashes establish correspondence of bytes; the mathematical conclusion comes from the verification chain applied to the identified data.

## Exercises and audit questions

**Exercise 23.1.** *An FFT bug changes a cached first derivative. Why can the full chain still be valid if it reports success?*

*Answer.* The cached field is merely a conforming candidate. The C verifier computes its residual and flux mismatch for the prescribed exact direction. A poor candidate normally widens the enclosure or causes failure; it is not assumed to solve the intended PDE.

**Exercise 23.2.** *The entry pass gives an enclosure centred at  $c$  with radius  $r$ . The mode file stores a decimal  $c_0$ . Is using radius  $r$  around  $c_0$  sufficient?*

*Answer.* In general no: the difference between  $c_0$  and the entry centre must be covered as well. The second C pass proves containment of the whole internal enclosure in the ball built from the actual mode file, and also checks the conjugate entry.

**Exercise 23.3.** *Why are four exact zeros compatible with strict local maximality?*

*Answer.* They are precisely the tangent directions to similarities, along which  $J/A^2$  is constant. On a local slice fixing translation, rotation, and scale, the Hessian is strictly negative definite.

**Exercise 23.4.** *What follows if `RESULT.json` says `NOT_CERTIFIED` at  $m = 64$ ?*

*Answer.* This chain at this mesh and these candidates has not proved the required signs. It supplies no counterexample to local maximality. The successful  $m = 128$  checks for  $n = 19, 20$  illustrate the difference between a failed enclosure test and a false mathematical claim.

## References

- [1] C. Bernardi, M. Dauge, and Y. Maday. Compatibilité de traces aux arêtes et coins d'un polyèdre. *Comptes Rendus de l'Académie des Sciences, Série I, Mathématique*, 331(9):679–684, 2000.
- [2] B. Bogosel and D. Bucur. On the polygonal Faber–Krahn inequality. *Journal de l'École polytechnique—Mathématiques*, 11:19–105, 2024.
- [3] B. Bogosel and D. Bucur. Polygonal Faber–Krahn inequality: local minimality via validated computing, 2024. arXiv:2406.11575.
- [4] P. Grisvard. *Elliptic Problems in Nonsmooth Domains*, volume 24 of *Monographs and Studies in Mathematics*. Pitman, Boston, 1985.
- [5] A. Laurain. Distributed and boundary expressions of first and second order shape derivatives in nonsmooth domains. *Journal de Mathématiques Pures et Appliquées*, 134:328–368, 2020.
- [6] Q. Li and X. Liu. Explicit finite element error estimates for nonhomogeneous Neumann problems. *Applications of Mathematics*, 63(3):367–379, 2018.
- [7] X. Liu and S. Oishi. Verified eigenvalue evaluation for the Laplacian over polygonal domains of arbitrary shape. *SIAM Journal on Numerical Analysis*, 51(3):1634–1654, 2013.
- [8] S. Repin. *A Posteriori Estimates for Partial Differential Equations*, volume 4 of *Radon Series on Computational and Applied Mathematics*. Walter de Gruyter, Berlin, 2008.
- [9] G. J. Tee. Eigenvectors of block circulant and alternating circulant matrices. *New Zealand Journal of Mathematics*, 36:195–211, 2007.
- [10] The FLINT development team. Arb: arbitrary-precision real numbers, FLINT documentation. <https://flintlib.org/doc/arb.html>, 2026. Accessed 13 September 2026. Recorded certification runs identify their own FLINT version.